

```

clc; clear; close all;
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
Nt = 2; Nr = 2;
BER = zeros(1, length(SNR));
for i = 1:length(SNR)
    bits = randi([0 1],10000,1);
    symbols = 2*bits-1;
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
    tx = H(1,1)*symbols' + noise(1,:);
    rx = real(tx) > 0;
    BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
subplot(2,1,1)
semilogy(SNR_dB,BER, '-o', 'LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
subplot(2,1,2)
plot(SNR_dB,1-BER, '-s', 'LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on

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